MILITARY SPECIFICATION

WELDING MACHINE, ARC, DIRECT-CURRENT, VARIABLE VOLTAGE, PORTABLE, (SHIPBOARD USE)

1. SCOPE

1.1 This specification covers a shockproof, portable, welding machine, direct-current (d.c.) motor generator design, for shipboard service.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-C-915	- Cable, Cord and Wire, Electrical (Shipboard Use).
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-D-963	- Drawings, Electrical, Hull and Mechanical Equipment For Naval Shipboard Use.
MIL-C-1213	- Cups, Grease.
MIL-C-2212	- Controllers, Alternating-Current, Naval Shipboard.
MIL-G-3111	- Generators, Electric, Direct-Current (Naval Shipboard Use).
MIL-E-16298	- Electric Machines Having Rotating Parts and Associated Repair Parts; Packaging Of.
MIL-I-16910	- Interference Measurement, Electromagnetic, Methods and Limits.
MIL-M-17060	- Motors, Alternating Current, Integral Horsepower (Shipboard Use).
MIL-E-17555	- Electronic and Electrical Equipment and Associated Repair Parts, Preparation For Delivery Of.
MIL-G-18473	- Generators and Motors, Direct Current, Naval Ship Propulsion.
MIL-G-18709	- Grease, Ball and Roller Bearing.
MIL-E-22200/1	- Electrodes, Welding, Mineral Covered, Iron-Powder, Low-Hydrogen Medium and High Tensile Steel, As Welded Or Stress-Relieved Weld Application.

STANDARDS

MILITARY

MIL-STD-167	- Mechanical Vibrations of Shipboard Equipment.
MIL-STD-248	- Qualification Tests For Welders (Other Than Aircraft Weldments).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 3431

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OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York, N.Y. 10016.)

3. REQUIREMENTS

3.1 Qualification.- The variable-voltage, direct-current, electric arc welding machine furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Basic requirements.- Except as otherwise specified herein, the welding machine shall conform to MIL-E-917.

3.2.1 Material.- The material shall be in accordance with MIL-E-917.

3.3 Design.-

3.3.1 Apparatus comprising a machine.- The welding machine shall be completely assembled, compactly mounted, and shall comprise the equipment necessary for furnishing electric energy to the welding circuit; namely, a motor with suitable starter, a generator with provision for current (see 6.2.8) adjustment, and exciter (if required), suitable reactance, resistance and connectors for the welding-circuit leads. A dynamotor may be used in lieu of a motor-generator.

3.3.2 Weights and dimensions.- The welding machine shall be as compact and lightweight as practical, consistent with strength and reliability, but shall not exceed 400 pounds. The overall dimensions shall be such as to allow the machine to pass through a 25-inch diameter hatch.

3.3.3 Safety.- Particular care shall be taken in the design and construction of the welding machine to insure safety to the operating personnel from danger of electrical shock or moving parts.

3.3.4 Suitability of operation.- The welding machine shall be suitable for welding with coated metallic electrodes conforming to MIL-E-22200/1. The characteristics of the set shall be such that an operator qualified in accordance with MIL-STD-248 may easily strike and maintain a stable arc and satisfactorily weld at any setting (see 6.2.1) within the specified range of the machine under normal conditions without losing the arc more than once per minute.

3.3.5 Power supply.- The welding machine shall operate from a power source of either 440 or 220 volts, 60 cycles, 3 phase, alternating current (a.c.), and shall be readily reconnectable for either voltage (see 6.2.9). Unless otherwise specified (see 6.1), the welding machine shall be connected for use on a 440 volt power supply prior to shipment.

3.3.6 Output current.- The welding machine shall have the range of output current shown in table I.

Table I - Range of output current.

Size	Rating of generator (see 3.3.19.2)		Range of output current: 20 volts to rated voltage
	Rated normal current amperes (see 6.2.5)	Rated voltage	
200	200	30	40-250

Note.- Ratings shall be based on continuous welding at rated load on a 60 percent duty cycle.

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3.3.7 Operating characteristics. -

3.3.7.1 Current adjustment. - Current within the range specified in table I shall be obtained at any voltage between 20 volts and rated voltage (see 6.2.4) inclusive. There shall be no setting at which it will be difficult to weld. Current shall be adjustable between the minimum and maximum values specified in table I in not less than 30 increments. Not more than two controls shall be used for current adjustment. At rated voltage, the increase in current from any setting to the next higher setting shall not exceed 5 percent of the current at the lower setting.

3.3.7.2 At a setting corresponding approximately to rated output (see 6.2.6) (1) starting cold and (2) starting at the conclusion of a 4-hour run at no-load, the current shall vary not more than plus or minus 15 percent from the initial current during a 10-minute run against a resistive load adjusted to maintain rated voltage. The adjustments on the welding machine shall not be changed during this run.

3.3.7.3 Starting at a machine temperature corresponding to that at the conclusion of a full-load heat run, the machines shall operate satisfactorily at the maximum current obtainable at the lowest applicable arc voltage (see 6.2.10) without damage to the insulation or other parts for 2 minutes out of each 4 minutes for 10 minutes total elapsed time. The overload protective device may be rendered inoperative during this run.

3.3.7.4 The maximum open-circuit voltage (see 6.2.7) shall not exceed 100 volts under any operating conditions.

3.3.7.5 Volt-ampere characteristics. - Volt-ampere characteristic curves (see 6.2.2) shall show that at all points, below rated voltages, the voltage decreases with an increase in current. No point of a curve shall indicate a current higher than that for short-circuit. The machine shall have such characteristics that for any setting, at any arc voltage within the range of values specified in table II, a 10 percent change in arc voltage shall not result in more than a 12 percent change in current.

Table II - Range of arc voltage.

Amperes	Arc voltage
25 to 50	18 to 25
50 to 100	18 to 27
100 to 150	20 to 29
150 to 200	21 to 30
200 to 300	22 to 33
300 to 400	23 to 38
400 to 500	25 to 40

3.3.7.6 Reactance. - The welding machines shall include reactance in the welding circuit effective for all welding conditions. The reactance may be internal or external, or both. External reactance shall be suitably protected, dripproof, and secured by rugged connections.

3.3.8 Short-circuit and recovery characteristics. - The welding machine shall conform to the following short-circuit and recovery characteristics. The data shall be determined by an oscillograph.

3.3.8.1 Short-circuit current. -

3.3.8.1.1 When starting with open-circuit and then shorting the welding terminals, the peak or momentary short-circuit current for any setting shall not exceed 500 percent of the normal current for that setting.

3.3.8.1.2 When starting with normal current and then shorting the welding terminals, the peak or momentary short-circuit current for any setting shall not exceed 400 percent of the normal current for that setting.

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3.3.8.1.3 The steady-state short-circuit current for any setting shall be not greater than 200 percent nor less than 110 percent of the normal current for that setting.

3.3.8.2 Recovery voltage.- On any setting, the minimum recovery voltage, when a short-circuit is opened through a resistance giving normal load (see 6.2.3) for that setting, shall be not less than 70 percent of the final load voltage on that setting.

3.3.8.3 The current shall not drop below 60 percent of the normal current for any setting when a short-circuit is opened through a normal load for that setting.

3.3.8.4 Time of recovery.- The permissible time of recovery from short-circuit current to within 5 percent of normal current for any setting shall not exceed 0.5 second.

3.3.9 Ambient temperatures.- The welding machine shall be designed for an ambient temperature of 40°C.

3.3.10 Maximum permissible temperature rises.- The temperature rises of the welding machine shall not exceed the values shown in table III.

Table III - Maximum permissible temperature rises.

Item	Parts of control equipment, motor generator, and exciter to be measured	Method of measurement	Class A Insulation of MIL-E-917	Class B Insulation of MIL-E-917
			°C	°C
1	Single layer insulated windings or uninsulated windings in contact with insulation.	(1) Thermometer	55	75
2	Multiple layer armature and field windings.	(1) Thermometer	50	70
		(2) Resistance --	60	80
3	Bare copper not in contact with insulation.	(1) Thermometer	80	80
4	Commutators with connections to insulated windings.	(1) Thermometer	65	85
5	Bearings ----------	(1) Thermometer	40	40
6	Core or other mechanical parts in contact with insulated parts.	(1) Thermometer	50	70
7	Contacts:	(1) Thermometer		
	Laminated copper contacts ----------	----------	50	50
	Solid contacts --------	----------	60	60
	Solid contacts, silver faced ----------	----------	75	75

Note.- The temperature rise for item 2 shall be measured by method 1 (thermometer) and method 2 (resistance). The determination of the values to govern as obtained by these methods shall be taken as follows:

Less than 1 ohm. . . . Thermometer temperatures (method 1).
Over 1 ohm . . . . . Highest values of method 1 (thermometer) and method 2 (resistance).

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3.3.11 Commutation.- The motor, generator, and exciter shall operate at all loads between minimum and maximum current output with sparking conditions in accordance with MIL-G-18473.

3.3.12 Efficiency.- The efficiency of the equipment shall be not less than the applicable values shown in table IV, and shall be determined in the manner described in 4.5.10. The power factor for alternating-current motors shall be not less than that shown in table IV.

Table IV - Efficiency values.

Percentage of normal rated output of generator	200-ampere at 30 volts	
	Efficiency	Power factor
	Percent	Percent
30	40	47
50	47	58
75	49	70
100	47	78
125	45	80

3.3.13 Dielectric strength.- The insulation of the welding machine shall be designed to withstand a dielectric test voltage of twice the normal voltage of the circuit plus 1,000 volts.

3.3.14 Balance.- The welding machine shall run well balanced, both electrically and mechanically, in accordance with the applicable motor and generator specifications, at all loads and voltages within the service range.

3.3.15 Parallel operation.- The welding machine shall operate satisfactorily in parallel with other machines of the same type. If it is necessary that equalizer connections be used, suitable readily accessible equalizer terminals shall be provided with proper markings.

3.3.16 Shock.- The welding machine shall withstand the high impact (H.I.) shock tests specified in 4.2.1. Grade A, class I equipment in accordance with MIL-S-901 is required.

3.3.17 Vibration.- Welding machines shall be designed and constructed to withstand severe vibrations transmitted through the ship's structure in the course of normal operation of the vessels. These vibrations may ordinarily be expected to vary in frequency from a very low value up to approximately 1,350 cycles per minute and have a double amplitude of as much as 0.030 inch; however, under special circumstances, these values may be exceeded.

3.3.18 Electromagnetic interference.- The equipment shall be designed to insure a minimum amount of electromagnetic interference in accordance with MIL-I-16910.

3.3.19 Classification of components.-

3.3.19.1 Alternating-current motors.- Alternating-current motors shall conform to MIL-M-17060 and the following:

Volts	Unless 220 volts is specified (see 6.1), furnish connected for 440 volts.
Phase	3
Cycles	60
Duty rating	Full load rating, as specified in 3.3.6, no load, continuous.

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Enclosure	Dripproof protected.
Speed	Constant.
Starting torque	Normal.
Starting current	Low.
Bearings	Ball or roller.
Service	A.

3.3.19.2 Generators.-

3.3.19.2.1 Welding generators.- Welding generators shall conform to MIL-G-3111, and shall be of the following classification:

Voltage	Variable, as specified in 3.3.6.
Current	As specified in 3.3.6.
Duty rating	Varying: full-load rating, as specified in 3.3.6, no load, continuous.
Enclosure	Dripproof protected.
Bearings	Ball or roller.

3.3.19.2.2 Exciter generator.- Exciter generators, when used, shall be continuous duty, and shall conform to MIL-G-3111.

3.3.19.3 Motor starters.- Alternating-current motor starters shall be in accordance with MIL-C-2212 and the following classification:

Enclosure	Dripproof.
Rating	As required for motor.
Operation	Magnetic.
Type	Across-the-line.
Function	Motor starting.
Duty	*Continuous.*
Protection	Undervoltage protection (LVP) and overload protection (hand or automatic reset) are required. Unless otherwise specified in the contract or order, emergency-run feature and short circuit protection shall not be furnished.
Performance	Nonautomatic, manually operated type.
Service	A (see 3.3.19.1).
Ambient	As specified in 3.3.9.
Master switch	*Manual, local, pushbutton type.*

3.3.20 Bearings.- Manually greased or permanently sealed ball bearings shall be used.

3.3.20.1 Lubrication.- The lubrication of ball bearings shall conform to MIL-G-18709 unless bearings of the sealed permanently lubricated antifriction type are used in which case requirements of this paragraph and paragraphs 3.3.20.1.1 and 3.3.20.1.2 shall be inapplicable.

3.3.20.1.1 Particular care shall be taken to provide a method of lubrication which will prevent leakage of oil or grease along the shaft and into the electrical windings.

3.3.20.1.2 Where grease lubrication is provided, the lubricant shall be applied by means of compression grease cups conforming to type I of MIL-C-1213. Drain plugs shall be supplied in the bearing housing, located in the lowest part of the housing, and in the case of single shielded bearings, on the same side of the bearing as the filling connection. Extension grease pipes of suitable size shall be supplied for grease cups and drain plugs when needed to provide ready accessibility for these fittings.

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3.3.21 Mounting.- The motor and generator (or dynamotor) shall be mounted upon a substantial common metal base or provided with other satisfactory equivalent means of support. The motor-generator may be of either the two- or four-bearing type. The two-bearing type shall have a solid common shaft. The four-bearing type shall have the motor and generator shafts connected by a rugged flexible coupling. The mounting supports of the four-bearing type shall be doweled to preserve shaft alinement. Where an exciter is used, such additional bearings as necessary shall be provided. The exciter shall be connected by a flexible coupling, unless a solid common shaft is used for the motor or generator and exciter.

3.3.22 Canopy enclosure.- When specified (see 6.1), the machines shall be provided with a canopy enclosure so constructed that exposure to the weather will not interfere with the operation of the machine. The canopy shall have a removable rigid metal frame suitable for protecting the entire outfit, and shall be fitted on the sides and ends of the machine. The top of the canopy shall be galvanized iron, or galvanized steel, not less than 0.05 inch thick, with duck or galvanized sheet metal sides arranged to provide access to all parts of the equipment, and securely fastened in the closed and open positions. The top shall be supported by not less than four vertical steel angles not less than 2 by 2 by 1/4 inch (or equivalent), suitably tied, braced, and rigidly connected by angles of the same size. The removable connections shall be made with not less than two 5/8-inch bolts at each support. Duck shall weigh not less than 15 ounces per square yard, and shall be brown or khaki color.

3.3.23 Lifting means.- Machines provided with a canopy shall have not less than two lifting lugs or eyebolts of not less than 7/8-inch diameter wire, fastened to the canopy frame, suitable for lifting the set. Unless otherwise specified in the contract or order, machines without canopy shall have lugs, eyes, holes, handles, or hand rails suitable for lifting the machine.

3.3.24 Welding control.-

3.3.24.1 The welding control shall include necessary rheostats, contact-making devices, switches, meters, polarity indicators, and other equipment, as necessary, to provide adjustments for the welding range specified in 3.3.6. The equipment shall be mounted in a dripproof enclosure. The enclosure preferably shall include the motor control also in a compact form. The welding control shall be easily accessible at convenient height, and of such design that it can be readily operated by an operator wearing a welder's glove. Current-carrying parts shall be mounted on impregnated asbestos board, or insulated by plastic insulating material, in accordance with MIL-E-917, and supported by steel panels or framework except that asbestos board shall not be used. Welding terminals shall be protected from accidental contact or short circuit. All other live parts shall be enclosed. Welding terminals shall not be located in the same enclosure with terminals for the connection of power leads.

3.3.24.2 Calibration.- All current-adjusting devices shall be provided with suitably numbered or calibrated plates.

3.3.24.3 Polarity reversing switch.- The welding control shall include a switch for reversing the polarity of the welding terminals. The polarity shall not reverse automatically under normal operation when the generator is being operated singly or in parallel with another similar generator. Polarity reversing switches will not be required.

3.3.24.4 Generator terminal connections.- Welding terminals shall be the stud and hexagonal-nut type. The diameter of the stud type shall be not less than 1/2 inch.

3.3.24.5 Motor power connections.- A suitable polarized receptacle shall be mounted on the machine for supplying power to the motor. A plug for insertion in the motor power receptacle shall also be provided as a part of the equipment. The plug shall be suitable for attaching a motor power cable consisting of power leads and a grounding lead of ample capacity. The plug shall be of such design that when detached from the receptacle, live current-carrying parts are protected from shorting.

3.3.25 Motor supply cable.- Unless 220 volt connection is specified in the contract or order, 100 feet of motor supply cable, type SHOF, size 9 conductor, in accordance with MIL-C-915, shall be furnished. When 220 volt connection is specified, cable shall be type SHOF, size 4 conductor.

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3.3.26 Wiring. -

3.3.26.1 Internal wiring. - All permanent internal wires for control devices shall be of sufficient capacity for duty intended. The conductor may be either copper or steel core copper wire. All copper and steel core copper wire smaller than 10,000 circular mils shall be stranded, except that in the case of supported connecting leads where the length between supports does not exceed 18 inches, a solid copper or steel core copper wire greater than 4,000 circular mils shall be used. All connecting wires liable to chafing and grounding shall be well insulated: Such insulation shall conform to the standards of quality required in MIL-C-915. Where exposed to heat of resistor elements or operating coils, the insulation shall be of the heat and flame resistant type and unless otherwise specified in the contract or order, it shall conform to the insulation requirements for type SHFS wire of MIL-C-915.

3.3.26.2 External wiring. - External permanent wiring on the machine shall be enclosed in flexible metal conduits or clamped in position to prevent abrasion of the insulation.

3.3.26.3 A diagram of connections shall be attached to the inside of the control equipment cover and shall be protected by a transparent varnish or other suitable material.

3.3.27 The welding machine shall be provided with suitable truck, base, or running-gear supports. The bedplate, if used, shall be suitably bolted to the base or truck. If the base or truck frame is sufficiently rigid, the bedplate may be omitted. If suitable lugs or feet for supporting the skids are cast on or welded to the motor and generator frame, a truck frame will not be required. Unless other specified (see 6.1) the welding machine shall be equipped with skids in lieu of wheels. The skids, or wheels when specified, and running gear shall be so constructed as to support a load equal to four times the normal loading of the parts without breakage or permanent set in any parts. The wheels when specified shall be not less than 4.5 inches in diameter with not less than 1-1/4-inch thread. The front wheels shall be so constructed as to turn through 180 degrees so as to provide a small turning circle. A tongue of rigid construction shall be provided for pulling the machine by hand or truck. The machine shall be designed to operate satisfactorily when inclined at an angle of 15 degrees in any direction from the normal position. The machine shall also be inherently capable of remaining upright (that is, mechanically stable) when tilted 20 degrees from the normal position to either side.

3.4 Onboard repair parts. - Onboard repair parts and tools shall be based upon the total number of machines furnished for each vessel, and shall be supplied in accordance with table V as specified (see 6.1).

3.5 Designating and marking. -

3.5.1 Identification plates. -

3.5.1.1 Identification plates shall be attached to the part of the machinery or equipment which will not ordinarily be renewed during its normal service life, and shall be located in a readily accessible position where they can be read at all times without danger to personnel.

3.5.1.2 When an identification plate or designating plate is mounted on a switchboard or panel, it shall be placed in close proximity to the equipment to which it refers and generally either directly above or directly below it.

3.5.1.3 Particular attention shall be given to securing plates to sheet metal panels. Plates mounted on such panels shall be secured by machine screws, nuts, and lockwashers (or other satisfactory means of locking the nuts).

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Table V - Onboard repair parts.

Item	Parts applicable to the welding machines installed	Quantities to be furnished per vessel and carried on board: 1 to 4 machines per vessel	5 to 8 machines per vessel	9 or over machines per vessel
	For generator, exciter (if used):			
1	Bearings. . . . . sets	1	2	3
2	Commutator brushes . . . . . sets	3	6	8
3	Brush rigging insulation, each type . . . . . sets	1	1	1
4	Commutator brush holders, each type (sets complete with springs)	1	2	2
5	Commutator brush holder spring, each type number . .	3	4	6
6	Bearing lubricant seals . . . . . sets	3	4	5
7	Field coils of each size and type . . . . . number	1	1	1
	For motor:			
8	Bearing for motor end of unit	1	1	1
9	Bearing lubricant seals . . . . . sets	3	4	5
10	Stator coils, wedges and slot insulation material sets. .	1	1	1
	For controllers:			
11	Complete set of contacts	1	1	1
12	Complete set of springs.	1	1	1
13	Complete set of overload heaters	1	1	1
14	Shunt coil, each size and type.	1	1	1
15	Resistor unit, each size and type	1	1	1
16	Female-angle 1-hole couplings for generator terminals sets	1	2	2

Note 1. - When shunt and series coils are so constructed that they cannot be readily separated from each other, the unit will be considered as a repair part for both.
Note 2. - No series control coils, except heater coils, shall be furnished as repair parts.
Note 3. - When two or more identical elements of a pushbutton type master switch are furnished, only one element need be supplied as a repair part.

3.5.2 Identification plate marking. - The data marked on main identification plates shall include the following items:

(a) Manufacturer's name, identification symbols, serial number and contract number.
(b) Salient design characteristics: Welding generator nominal size (amperes), rating (amperes and volts), current range; motor horsepower, volts, frequency, number of phases, full-load amperes; specification number.
(c) Blank space for Government inspector's official stamping.
(d) Federal stock number.

3.5.3 Frame and shaft marking. - The manufacturer's serial number shall also be stamped in the solid metal of the machine frame covered by the identification plate.

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3.5.4 Direction of rotation.- A suitable plate or marking indicating the direction of rotation shall be permanently attached to the machine in a conspicuous location.

3.6 Drawings.- Drawings shall be classes A and B in accordance with MIL-D-963 and as specified herein.

3.6.1 Master drawings.- In addition to the contents required by MIL-D-963, master drawings shall include the following minimum data:

(a) Outline end view and side view (small scale) showing over-all dimensions, location, and method of securing of terminals, and weight.
(b) Assembly end and side view showing all parts of the generator, motor and exciter, if used. Parts shall be marked with part numbers and referred to the list of material. The assembly views shall be of sufficiently large scale to show details of construction. One quarter section (or more if necessary) of the end view shall show a section through poles and field coils of the generator. Also, one quarter section of the end view shall show a section through poles and field coils of the driving motor. A section through poles and field coils of the exciter, if provided, shall also be shown. The assembly view shall show a longitudinal section of the generator, motor and exciter, if used, above the center line and shall indicate the field coil connections. The size of wire, number of turns and method of securing the generator armature and exciter armature banding in place shall be shown.
(c) Complete winding data and insulation data for the field coils covered by notes with arrows directed to sectional views of the field coils.
(d) View of slot section of armature of generator, motor, and exciter showing the copper conductors and slot insulation in place and giving complete insulation data.
(e) Development of the generator and exciter armature winding, giving winding data. A partial diagram is sufficient for a uniform winding. For an armature having a nonuniform winding involving the use of dummy coils or dead commutator bars, a complete development or a partial development with a complete tabulation shall be shown. The size of copper used for all connections shall be shown.
(f) Detailed working drawing of the shaft.
(g) Assembly views of the generator and motor control showing location of parts and referring to the parts list.
(h) Complete wiring diagrams of the generator, motor, exciter, if used, and control.
(i) Manufacturer's machine data as shown on figure 1.

3.6.2 Certification data.- In addition to the contents required by MIL-D-963, certification data sheets shall include the following minimum data:

(a) Generator volts and amperes.
(b) Motor horsepower, volts, frequency, full-load current, and locked-rotor current.

3.6.3 Outline drawings.- Outline drawings shall conform to MIL-D-963, and shall contain the following minimum data:

(a) Outline: Overall and principal dimensions.
(b) Method of assembly and disassembly.
(c) Weights of complete unit, running gear if provided, complete rotating element and spares.
(d) Complete rating of generator and motor.
(e) Class of insulation.
(f) Maximum temperature rise.
(g) Ball bearings used; show the bearing manufacturer's name, the trade or catalog symbol, and the Military specification number including classification such as grade, or type, of the particular bearing or bearings used.
(h) List several alternate bearings of suitable rating which can be used.
(i) Complete list of repair parts.
(j) A definite statement as to whether or not any exceptions are being taken to applicable specifications referenced in 2.1, with a full description of such departures, if any.

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Outline drawings may consist of manufacturer's shop drawings, manuals, data sheets, catalogue pages, blueprints or other suitable prints. The manufacturer may, if he so desires, use copies of approved master drawings as outline drawings. Outline drawings shall not be furnished in a reproducible form. An exception is made, however, in the case of equipment requiring manuals in a reproducible form. In these instances the drawings made up for the manual (see 3.7) may also serve as outline drawings. Outline drawings may be of any reasonable dimensions provided the details listed above are clearly shown thereon.

3.7 Manuals.- The manufacturer's commercial off-the-shelf manual shall be provided. It shall include assembly, interconnection, set-up or installation, operation and repair parts of the welding machine. The manual shall contain all information necessary for trouble shooting, normal maintenance including lubricating instructions, and the data, in the format, shown on figure 1. The manual shall have substantial covers adequate for its protection with normal handling. A complete wiring diagram of the welding machine shall be provided. The number of manuals required shall be as specified (see 6.1).

3.8 Workmanship.- The equipment, including all parts and accessories, shall be manufactured in accordance with first class manufacturing practices in all respects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests.[1/]- Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified in 4.2.1 through 4.2.3 and 4.5.

4.2.1 Shock tests.- The welding machine shall be subjected to the type A, H.I. shock tests in accordance with MIL-S-901 for grade A, class I, equipment, and shall be made on the light-weight or medium-weight shock machine as determined by the weight of the completely assembled welding machine, including running gear.

4.2.1.1 Mounting.- The welding machine shall be tested complete with running gear if the unit can be satisfactorily secured to the shock machine; otherwise, the unit shall be tested exclusive of running gear. For tests on the light-weight machine, the unit shall be secured to the standard mountings shown in MIL-S-901 for bulkhead mounted, or deck or platform, mounted equipment, or modifications of these mountings satisfactory to the bureau or agency concerned. For tests on the medium-weight machine the welding machine shall be attached to the standard mounting platform. Bolts of sufficient strength and number shall be used to secure the welding machine to the appropriate mounting on the basis that the combined ultimate tensile strength of the bolts shall be at least equal to 50 times the weight of the welding machines.

4.2.1.2 Number of blows.- The number of blows shall be as specified in MIL-S-901. The welding machine shall be tested, on alternate blows, in each of its principal operating conditions; that is, running and standing. If practicable, the machine shall be tested in the running condition with the generator delivering rated voltage and current.

4.2.1.3 Inspection after shock test.- Upon completion of the shock test, the welding machine shall be carefully examined and tested by welding to determine the extent of any damage to the mechanical and electrical components.

4.2.1.3.1 Any noticeable increase or decrease in noise or vibration produced by the machine shall be investigated and its cause determined. However, if disassembly is required for this check, it shall be made during the inspection specified in 4.2.1.6.

[1/] Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3 and 6.3.1).

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4.2.1.4 The operating characteristics obtained in 4.5.3 and 4.5.4 shall be spot-checked to determine if they have changed by more than 10 percent.

4.2.1.5 The welding machine shall be operated to obtain approximately rated temperature rise of all components, then disconnected from its source of power and a dielectric strength test shall be made to determine compliance with 3.3.13, except that it shall be made with an applied voltage equal to 65 percent of that specified therein.

4.2.1.6 The welding machine shall be disassembled following the check test specified in 4.2.1.3.1 to 4.2.1.5 inclusive, and inspected thoroughly for damage. The extent of disassembly need be only to the point where the condition of the unit may be easily observed. The effects of the shocks and subsequent check tests on the structure, bearings and insulation shall be carefully observed and recorded.

4.2.1.7 Causes for rejection. - Failure of the following phases shall be cause for rejection of the welding machine under the H.I. shock tests:

(a) Breakage of any parts.
(b) Appreciable distortion or dislocation of any parts, including gear, poles, coils, brushes, and bearings.
(c) Nonconformity to the tests specified in 4.2.1.3.1 to 4.2.1.6, inclusive.
(d) Starting of the unit from the deenergized condition or stopping of the unit from the energized condition as a result of any shock blow.
(e) Damage to the extent that proper operation or control of the welding machine cannot be exercised.

4.2.2 Vibration. - The welding machine shall be tested in accordance with type I of MIL-STD-167 to determine compliance with 3.3.17.

4.2.3 Electromagnetic interference. - The welding machine shall be tested in accordance with MIL-I-16910 to determine conformance with 3.3.18.

4.3 Sampling for quality conformance inspection. -

4.3.1 Lot. - All welding machines presented at one time shall be considered a lot for purposes of inspection.

4.3.2 Sampling for group A examination and tests. - A random sample of welding machines shall be selected from each lot and subjected to each of the group A examination and tests specified in 4.4.1 with quality conformance based on the sampling requirements of table VI.

4.3.3 Sampling for group B tests. - A random sample of welding machines shall be selected from each lot in accordance with table VII, and subjected to each of the group B tests specified in 4.4.2.

4.3.4 Sampling for group C tests. - A sample number of welding machines in accordance with table VIII shall be selected from the total number on the contract and subjected to the group C tests specified in 4.4.3.

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Table VI - Sampling for group A examination and tests

Number of welding machines in inspection lot	Number of welding machines in sample	(Number of equipments nonconforming on any group A examination or test)	
		Acceptance number	Rejection number
8 or under	5	0	1
9 to 15	7	0	1
16 to 25	10	0	1
26 to 40	15	0	1
41 to 110	25	1	2
111 to 300	35	1	2
301 and over	50	2	3

Table VII - Sampling for group B tests

Number of welding machines in inspection lot	Number of welding machines in sample	(Number of equipments nonconforming on any group B test.)	
		Acceptance number	Rejection number
8 or under	3	0	1
9 to 15	5	0	1
16 to 40	7	0	1
41 to 110	10	0	1
111 to 300	15	1	2
301 and over	25	2	3

Table VIII - Sampling for group C tests

Number of welding machines on the contract	Number of welding machines for group C tests
110 and under	1
111 to 500	2
501 and over	3

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4.4 Quality conformance inspection. -

4.4.1 Group A tests. - Each of the sample welding machines selected in accordance with table VI shall be subjected to each of the tests and examination specified below, and the results of each test and examination compared with specification requirements. Failure to conform to the specification requirements for any group A test and examination shall be counted as a defect and the welding machine shall not be offered for delivery. If the number of such nonconforming welding machines in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall not be offered for delivery.

Group A examination and tests	Reference paragraph
General examination	4.5.1
Measurement of resistance	4.5.2
Commutation	4.5.7
Mechanical balance	4.5.8
Insulation resistance	4.5.13
Dielectric tests	4.5.14

4.4.2 Group B tests. - Each of the sample welding machines selected in accordance with table VII shall be subjected to each of the tests specified below, and the results of each test compared with specification requirements. Failure to conform to the specification requirements for any group B test shall be counted as a defect and the welding machine shall not be offered for delivery. If the number of such nonconforming welding machines in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall not be offered for delivery.

Group B tests	Reference paragraph
Range of output current	4.5.3
Current adjustment measurement	4.5.4
Weight	4.5.11

4.4.3 Group C tests. - Each of the sample welding machines selected in accordance with table VIII shall be subjected to each of the tests specified below and the results of each test compared with specification requirements. In the event of any failure to conform to the specification requirements for any group C test the supplier shall correct the cause of failure on future production units.

Group C tests	Reference paragraph
Practical welding tests	4.5.5
Temperature rise tests	4.5.6
Short-circuit and recovery tests	4.5.9
Efficiency measurement	4.5.10
Inclined operation	4.5.12
Parallel operation	4.5.16

4.5 Examination and tests. -

4.5.1 Examination. - Each welding machine shall be subjected to a thorough examination to ascertain that the workmanship and design conform to this specification. The fit of parts shall be observed with particular reference to the interchangeability of such as are likely to require replacement during the normal service life of the unit.

4.5.2 Measurement of resistance. - The resistance of the armature and all fields and the temperature at which they are measured shall be taken and recorded in the test report.

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4.5.3 Range of output current.- Tests shall include taking sufficient data to plot volt-ampere curves at the 4 settings corresponding to maximum specified current at rated voltage, rated current at rated voltage, minimum specified current at 25 volts, and minimum specified current at 20 volts. All measurements taken shall be recorded in the test report.

4.5.4 Current adjustment measurements.- With a variable resistance load continually adjusted to maintain rated voltage at the machine terminals, measure the current output at each machine setting. Record the measured values together with the machine settings in the test report.

4.5.5 Practical welding tests.- Tests to determine suitability of operation shall be conducted by a qualified arc welding operator in accordance with MIL-STD-248 and shall include welding in all positions with 1/8- and 3/16-inch electrodes (200-ampere size), to determine compliance with 3.3.4 and 3.3.7.1.

4.5.6 Temperature rise tests.-

4.5.6.1 Assembly of welding machine.- The temperature rise tests shall be made only on the completely assembled welding machine.

4.5.6.2 Conditions of tests.- The temperature rises shall be determined under the following conditions:

(a) Operation of the machine continuously on the full-load setting with a load equal to the rated output current multiplied by the square root of the specified duty cycle. The heat run shall be continued until all temperature rises are ultimate.
(b) Operation of the machine without welding load but at maximum open circuit voltage on the generator (maximum generator excitation) for a period of 4 hours.
(c) Operation of the machine continuously with 20 volts at the terminals on the maximum current obtainable on the lowest current tap multiplied by the square root of the specified duty cycle. The heat run shall be continued until all temperature rises are ultimate.

4.5.6.3 Measurement of the ambient temperature.- The welding equipment may be tested at any convenient room temperature above 10°C. The temperature rise for any set of operating conditions shall be considered independent of ambient temperature between ambient temperatures of 10° and 50°C.

4.5.6.3.1 The ambient or room temperature shall be measured at two or more different points around and on a level with the welding equipment and at a distance of 3 to 6 feet therefrom. The temperature sensing elements shall be inserted in oil-filled cups of not less than 1-inch external diameter and 2 inches high, and shall be protected from drafts and from abnormal heat radiation. The value to be adopted for the ambient or room temperature during the tests shall be the mean of the temperature readings (as stated) taken at four equal intervals of time during the last quarter of the duration of the test.

4.5.6.3.2 The conditions in the testing room shall be such that the room temperature will not vary greatly during tests. A variation of 10°C., or over, during a period of 6 hours, or a proportional change for runs of shorter duration, shall in no case be exceeded. If the room temperature is very irregular during the run, or changes rapidly at the end, the test shall be repeated.

4.5.6.4 Methods of temperature measurement.- Temperature shall be measured in accordance with either method 1, 2, or 3 of MIL-E-917.

4.5.7 Commutation.- The observation of commutation shall be made in conjunction with and during the progress of other tests and at varying loads throughout the service range, including the 25 percent overload point. Also, at shutdown, the armature shall be rotated slowly by hand and any irregular markings or lack of uniform coloration of polish on the surface of the commutator covered by the brushes shall be carefully noted.

4.5.8 Mechanical balance.- The mechanical balance of the completely assembled welding machine shall be observed and when the machine is operated at rated current and voltage and at maximum no-load speed.

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4.5.9 Short-circuit and recovery tests. - Current-time relationship for these tests shall be determined by means of oscillograms. Starting with open-circuit, short-circuit the welding terminals by means of a contactor with blow-out coil. After the short-circuit has become steady, open the short-circuit through a resistive load which will give rated current for that setting. The peak short-circuit current shall also be determined when starting from normal load conditions. The time of recovery shall be the time elapsed from the opening of the short-circuit until the current is within 5 percent of the normal for that setting. Simultaneous machine voltage and amperage oscillograms shall be taken when a steady state short-circuit current is opened through a resistive load which will give normal load at that setting. Short-circuit and time of recovery characteristics, as specified in 3.3.8 shall be determined for at least the following settings:

(a) The setting corresponding to rated output of the generator.
(b) The setting corresponding to minimum specified output current at 20 volts (see 3.3.6).
(c) At least one other setting within the specified range (see 3.3.6). During these tests the oscillograph film shall travel at sufficient speed to distinctly show the maximum and minimum voltages and currents. All measurements taken shall be recorded in the test report.

4.5.10 Efficiency measurement. - The input shall be measured at the motor terminals at rated voltage, speed, and rated frequency. The power input shall include all power taken from an external source for excitation and motor drive. The output shall be measured at the welding terminals when the machine is delivering the loads specified in 3.3.12. The efficiency measurements shall be made immediately after the temperature run at rated output while the machine is at normal operating temperature. The percent efficiency shall be 100 times the measured output divided by the measured input. All measurements shall be recorded in the test report.

4.5.11 Weight. - The weight of the welding machine, complete with all component parts, shall be taken and recorded in the test report.

4.5.12 Inclined operation. - The welding machine shall be tested by inclining the machine to an angle of 15 degrees from its normal position in each of four positions, 90 degrees apart, along the axis of symmetry.

4.5.12.1 Under each of the four positions of inclination, the welding machine shall be run at no-load, but full speed, for a period of not less than one-half hour. During the progress of these tests the mechanical balance shall be as good as it was in the normal position. There shall be no pounding or grinding the bearings, and lubrication shall be satisfactory.

4.5.13 Insulation resistance. - This test shall be measured before the dielectric tests (see 4.5.14). Prior to the application of the test voltage, the windings shall be thoroughly discharged by connecting the windings to the frame. Separate measurements shall be made on the motor, generator, and controller windings except where these windings are internally connected. Circuits of equal voltage above ground shall be connected together. Circuits or groups of circuits of different voltage above ground shall be separated.

4.5.14 Dielectric tests. - The dielectric tests shall be made after all the above tests have been completed. This test shall be made upon the completely assembled unit.

4.5.14.1 Applied potential. - The test voltage shall be applied successively between each winding and all other windings and the grounded frame.

4.5.14.1.1 Test voltage. - The frequency of the test voltage shall be not less than 60 cycles. It shall approximate a true sine wave. The test voltage as specified in 3.3.13 shall be applied continuously for a period of 1 minute.

4.5.14.1.2 Measurement of test voltage. - The voltmeter method shall be used. In measuring the voltage with a voltmeter the instrument shall derive its voltage from the high voltage circuit either directly or through an auxiliary ratio transformer or by means of a voltmeter coil placed in the testing transformer.

4.5.15 No-load running input. - The no-load power input at rated voltage shall be measured and recorded.

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4.5.16 Parallel operation. - Tests to determine suitability for parallel operation shall be conducted by a qualified arc welding operator (see 4.5.5).

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Welding machine. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier, at the lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, supplier's name and destination.

5.1.2 Onboard repair parts. - Onboard repair parts shall be preserved and packaged level A, packed level C and marked levels A and C, respectively, in accordance with the requirements of MIL-E-17555 and MIL-E-16298 as applicable.

5.2 Domestic shipment and storage or overseas shipment. - The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.)

5.2.1.1 Preservation and packaging. -

5.2.1.1.1 Welding machine. - The welding machine shall be preserved and packaged in accordance with the level A requirements of MIL-E-16298 or MIL-E-17555 as applicable. When the size and mounting arrangement of the welding machine does not permit enclosure of the generator within a watervaporproof barrier as required in a method II package, the alternate method of packaging specified in MIL-E-16298 may be applied.

5.2.1.1.2 Repair parts. - Repair parts shall be preserved and packaged in accordance with the level A requirements of MIL-E-17555 and MIL-E-16298 as applicable.

5.2.1.2 Packing. -

5.2.1.2.1 Welding machine. - The welding machine shall be anchored, blocked, braced and cushioned and enclosed in a sheathed crate conforming to the requirements of MIL-C-104 and appendix thereto.

5.2.1.2.2 Repair parts. - Repair parts shall be packed level A in accordance with MIL-E-17555 and MIL-E-16298 as applicable. Unless otherwise specified in the contract or order, repair parts shall be packed separately and shipped concurrently with the basic equipment.

5.2.1.3 Unpacking instructions. - In addition to any special marking required by the contract or order, unpacking instructions of equipment shall be provided to prevent possible damage during removal. When practical, one set of these instructions shall be placed in a sealed waterproof envelope prominently marked

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"Unpacking Instructions" and securely affixed to the outside of the shipping container in a protected location. If the instructions cover a set of equipment packed in multiple containers, the instructions shall be affixed to the number one container of the set.

5.2.1.4 Manuals.- Manuals shall be packaged and packed in accordance with MIL-M-15071.

5.2.1.5 Marking.- Interior packages and exterior shipping containers shall be marked for shipment in accordance with MIL-STD-129.

5.2.1.5.1 Special marking.- Shipping containers shall be stencilled as follows:

"CAUTION-THIS EQUIPMENT MAY BE DAMAGED UNLESS UNPACKING INSTRUCTIONS ARE CAREFULLY FOLLOWED: UNPACKING INSTRUCTIONS ARE LOCATED (state where located)."

When practical, this marking shall be applied adjacent to the identification marking on the side of the container.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When welding machine is to be furnished connected for 220 volts (see 3.3.5 and 3.3.19.1).
(c) When canopy enclosure is required (see 3.3.22).
(d) When wheels are required in lieu of skids (see 3.3.27).
(e) Repair parts required (see 3.4).
(f) Number of manuals required (see 3.7).
(g) Preservation, packaging, packing, and marking requirements other than those required by 5.1.

6.2 Definitions.-

6.2.1 Setting.- A setting is defined as a point of adjustment of the welding generator controls.

6.2.2 Volt-ampere curve.- The volt-ampere characteristic of any setting is a plot of the terminal voltage versus generator output current for that setting as an attached load is varied to draw output current from zero to short-circuit current.

6.2.3 Normal load.- Normal load for any setting is the current which will flow at that setting through a connected external resistive load adjusted to give rated voltage at the generator output terminals.

6.2.4 Rated voltage.- Rated voltage for any size of machine is the voltage specified for that size.

6.2.5 Rated normal current.- Rated current for any size machine is the load specified for that size.

6.2.6 Rated output.- Rated output for any size machine is the normal output when the machine is operating at rated voltage and rated normal current.

6.2.7 Maximum open-circuit voltage.- The maximum open-circuit voltage is the open-circuit voltage at the generator output terminals with the generator set for maximum excitation and with the motor operating at no-load speed corresponding to rated full-load speed.

6.2.8 Current.- Unless otherwise specified herein, current is the generator output current.

6.2.9 Voltage.- Unless otherwise specified herein, voltage is generator output terminal voltage.

6.2.10 Arc voltage.- Arc voltage refers to the voltage drop which exists across the arc during the welding operation. Arc voltage is dependent upon the type and size of electrode used, the current, and the length of the arc.

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6.3 With respect to products requiring qualification, awards will be made only for such products as have, *prior to the time set for opening of bids*, been tested and approved for inclusion in Qualified Products List QPL 24150, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, *D. C.*, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

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1. Identification.
   (a) *Manufacturer*
   (b) Application

2. Reference specification.
   The arc-welding machine described hereunder will conform strictly to MIL-W-24150, the issue in effect (date).

3. Reference drawings

Title drawing	Mfr's dwg. No.	Bureau or agency dwg. No.

4. Welder rating.
   (a) Volts (rated) _____
   (b) Amperes (rated) _____
   (c) Temperature rise °C. _____
   (d) R.P.M. _____
   (e) Open circuit volts (max.) _____
   (f) Welding current and voltage range.
   _____ amps. at _____ volts
   _____ amps. at _____ volts

5. Motor data.
   (a) Volts _____
   (b) Frequency _____
   (c) No. of phases _____
   (d) Full load amps. _____

6. Welder classification.
   (a) Prime mover _____
   (b) Class of insulation
   Generator _____
   Exciter _____
   Motor _____
   (c) Mounting _____
   (d) Degree of enclosure _____
   (e) Method of cooling _____
   (f) Ambient temp. of reference °C. _____

7. Weights.
   (a) Welder (Complete) _____
   (b) Field coils

	Gen.	Motor	Exc.
Shunt			
Cross or compound			
Interpole			

   (c) Spare rotating element _____

8. Efficiency (guaranteed).
   (a) At rated load _____
   (b) At 3/4 load _____
   (c) At 1/2 load _____

9. Bearings.
   (a) Type (show the bearing manufacturer's name, the trade or catalogue number, and the classification, such as grade, or type, from the applicable Military specification (include Military specification number), of the particular bearing or bearing used)
   (b) Lubrication (give classification, such as, grade or type from the applicable Military specification (include Military specification number)
   (c) List of bearings of identical size and rating that may be used

10. Poles.

	Gen.	Motor	Exc.
(a) Main			
1. Number			
2. Air gap (nominal)			
(b) Cross or compound			
1. Number			
2. Air gaps (nominal)			
(c) Commutating			
1. Number			
2. Air gaps (nominal)			

11. Armature.

	Gen.	Motor	Exc.
(a) Diameter			
(b) Banding			
(c) Laminations secured to shaft by			

12. Commutator.

	Gen.	Motor	Exc.
(a) Diameter			
(b) Length			
(c) Wearing depth			
(d) Insulation between bars			
(e) Thickness of insulation between bars			
(f) Insulation to ground on shaft			
(g) Thickness of insulation to ground on shaft			

13. Brushes.

	Gen.	Motor	Exc.
(a) Number of studs			
(b) Number of brushes per stud			
(c) Size of brush			
Length			
Width			
Thickness			
(d) Trade name of brushes			
(e) Classification, such as grade, or type, from the applicable Military specification (include number of Military specification).			
(f) Name of brush manufacturer.			

14. Exceptions to specifications.

Figure 1. - Manufacturer's machine data.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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